

# MANAGING THE INNOVATION COMMITTEE OF THE CYBERSECURITY CLUB

SOFTWARE ENGINEERING  
PROJECT

PROJECT ID: 1

GROUP: 3

**Dr.ashwag** \_\_\_\_\_

# STUDENT WORK

ARWA SULTAN ALSULAMI  
S444001818@ST.UQU.EDU.SA  
444001818

NOORA SAAD ALWUTHAINANI  
S444001562@ST.UQU.EDU.SA  
444001562

ROBA FAHAD ALMALKI  
S444003119@ST.UQU.EDU.SA  
444003119

JUMANAH ALRAWIS  
S444000736@ST.UQU.EDU.SA  
444000736

HEBA FAHD ALMATRAFI  
S444010917@ST.UQU.EDU.SA  
444010917

# CHAPTER 1

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## **BUSINESS REQUIREMENTS:**

FACILITATING THE INNOVATIVE  
COMMITTEE'S MANAGEMENT OF MEMBERS'  
IDEAS AND CALCULATING THEIR  
VOLUNTEER HOURS IN THE CYBERSECURITY  
CLUB

## CHAPTER 2

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### STOCKHOLDERS:

1-HAMIDA KHAN , HEAD OF THE  
INNOVATION AND DEVELOPMENT  
COMMITTEE OF THE CYBERSECURITY  
CLUB

EMAIL: S441017934@ST.UQU.EDU.SA

2-EMAD QUTB , VICE CHAIRMAN OF THE  
INNOVATION AND DEVELOPMENT  
COMMITTEE OF THE CYBERSECURITY CLUB  
EMAIL: S444003871@ST.UQU.EDU.SA

### REQUIREMENTS GATHERING METHODS :

INTERVIEW

### QUESTIONS :

1-WHAT ARE THE MOST PROMINENT  
PROBLEMS YOU FACE ?

2-FROM YOUR POINT OF VIEW ,  
WHAT IS THE SOLUTION ?

# CHAPTER 4

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## FUNCTIONAL REQUIREMENTS :

### 1. LOGIN.

1.1 THE SYSTEM ALLOWS THE USER TO ENTER THEIR ID ALONG WITH THEIR PASSWORD FOR THE LOGIN PROCESS.

1.2 THE SYSTEM WILL VERIFY THE ENTERED USER AGAINST THE DATABASE OF THE CYBERSECURITY CLUB.

1.3 THE SYSTEM GRANTS ACCESS TO USERS WHO HAVE BEEN CONFIRMED AS MEMBERS OF THE CLUB.

1.4 THE SYSTEM DENIES ACCESS TO USERS WHO ARE NOT LISTED IN THE DATABASE AS CLUB MEMBERS.

1.5 THE SYSTEM PERMITS SENDING A MESSAGE TO NON-REGISTERED USERS ABOUT THE REGISTRATION OPENING TIMES AS MEMBERS.

# CHAPTER 4

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## FUNCTIONAL REQUIREMENTS :

2. THE USER MUST BE ABLE TO INPUT IDEAS FOR ACTIVITIES AND EVENTS.

2.1 THE SYSTEM ENABLES THE USER TO SELECT THE DEPARTMENT TO WHICH THEY WANT TO SUBMIT THEIR IDEA. THESE DEPARTMENTS INCLUDE THE ENTERTAINMENT IDEAS DEPARTMENT, SCIENTIFIC IDEAS DEPARTMENT, AND COLLABORATIONS DEPARTMENT.

2.2 THE SYSTEM ALLOWS THE USER TO WRITE THEIR IDEA ALONG WITH A DETAILED DESCRIPTION.

2.3 THE SYSTEM IS CAPABLE OF STORING THESE IDEAS IN THE SOFTWARE'S DATABASE.

2.4 THE SYSTEM ALLOWS FOR THE SUBMISSION OF THE IDEA TO THE CLUB'S SUPERVISOR FOR REVIEW.

# CHAPTER 4

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## FUNCTIONAL REQUIREMENTS :

3. ORGANIZING IDEAS ACCORDING TO PRIORITY.

3.1 ALLOWING THE SYSTEM ACCESS TO THE IDEAS STORED IN THE SOFTWARE'S DATABASE.

3.2 ALLOWING THE SYSTEM TO PRIORITIZE IDEAS BASED ON THE GLOBAL EVENT CALENDAR.

3.3 ALLOWING THE SYSTEM TO SHOW THE ORGANIZED IDEAS TO THE SUPERVISOR.

# CHAPTER 4

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## FUNCTIONAL REQUIREMENTS :

4. ALLOWING THE SUPERVISOR TO SEARCH FOR IDEA.

4.1 THE SYSTEM ALLOWS THE SUPERVISOR TO SPECIFY THE DEPARTMENT FROM WHICH THEY WANT THE IDEA.

4.2 THE SYSTEM DISPLAYS IDEAS FROM THIS DEPARTMENT ACCORDING TO PRIORITY.

4.3 THE SYSTEM ALLOWS THE SUPERVISOR TO SELECT ONE OF THE IDEAS.

4.4 THE SYSTEM DISPLAYS THE IDEA TO THE SUPERVISOR ALONG WITH ITS DETAILS.

4.5 THE SYSTEM DISPLAYS INFORMATION ABOUT THE IDEA'S ORIGINATOR TO THE SUPERVISOR.



## CHAPTER 4

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### FUNCTIONAL REQUIREMENTS :

5. SUPERVISOR CAN ACCEPT OR REJECT THE ACTIVITY IDEA.

5.1 THE SYSTEM ALLOWS THE SUPERVISOR TO REVIEW THE SUBMITTED IDEAS, ORGANIZED ACCORDING TO PRIORITY.

5.2 THE SYSTEM ENABLES THE SUPERVISOR TO REJECT OR APPROVE AN IDEA.

5.3 THE SYSTEM PERMITS THE SUPERVISOR TO PROVIDE A REASON WHEN REJECTING AN IDEA.

5.4 UPON APPROVAL BY THE SUPERVISOR, THE SYSTEM FACILITATES THE FORWARDING OF THE IDEA TO THE DEAN'S OFFICE.

5.5 THE SYSTEM ALLOWS FOR SENDING A MESSAGE TO THE DEAN'S OFFICE DETAILING THE REASON FOR APPROVAL ALONG WITH THE FULL PARTICULARS OF THE IDEA.

5.6 THE SYSTEM ENABLES THE DISPATCH OF A REJECTION OR APPROVAL MESSAGE, ACCOMPANIED BY THE RATIONALE, TO THE MEMBER WHO SUBMITTED THE IDEA.

# CHAPTER 4

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## FUNCTIONAL REQUIREMENTS :

### 6. SCHEDULING

6.1 THE SYSTEM ALLOWS THE SUPERVISOR TO WRITE CONFIRMED EVENTS ACCORDING TO THE AGREED-UPON DATES IN THE SOFTWARE'S INTERNAL CALENDAR.

6.2 THE SYSTEM PERMITS SENDING A MESSAGE TO ALL MEMBERS CONTAINING THE EVENT ALONG WITH THE AGREED-UPON DATE.

### 7. REMINDING MEMBERS AND SUPERVISOR OF THE EVENT DATE.

7.1 THE SYSTEM ALLOWS FOR SENDING NOTIFICATIONS TO THE SUPERVISOR REGARDING THE IDEA SUBMITTED BY THE MEMBER.

7.2 THE SYSTEM ENABLES SENDING A REMINDER MESSAGE TO THE MEMBERS 48 HOURS BEFORE THE EVENT.

# CHAPTER 4

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## NON-FUNCTIONAL REQUIREMENTS :

### **1-Availability**

1.1 The system must be available 24 hours a day for 5 days, So that the rest of the days of the week work on updating the updating the event schedule

### **2-Security**

2.1 Adding two step verification and encrypting all members and Presidents information should keep the system safe

### **3-Maintenance**

3.1 The system must update member information at the end of each semester

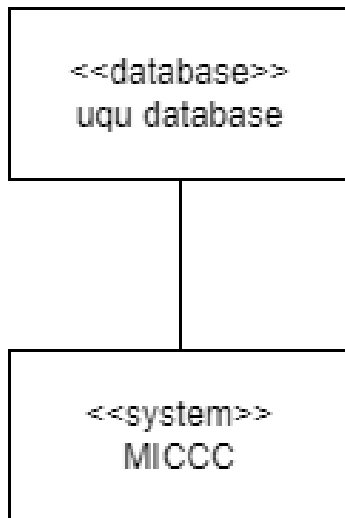
### **4-Usability**

4.1 use a simple interface to be friendly with With members and presidents

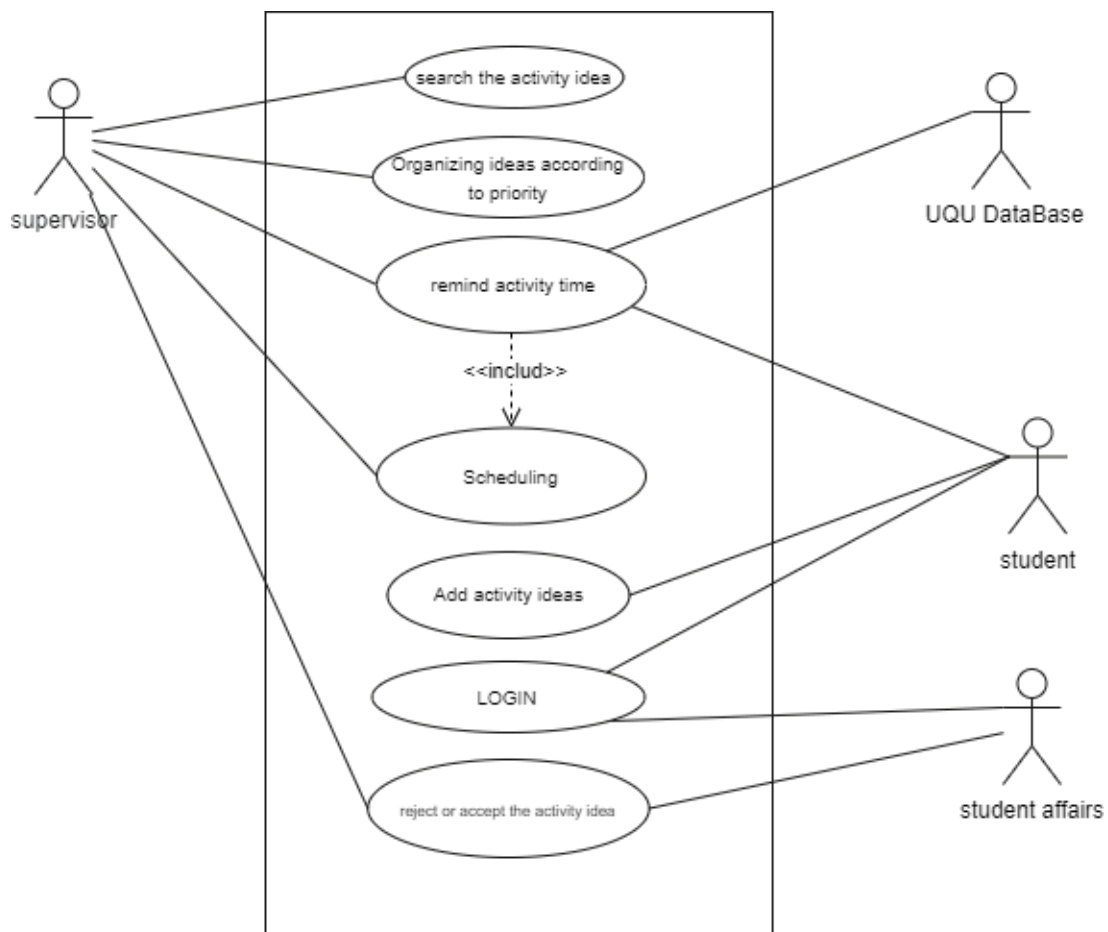
# CHAPTER 5

## UML SYSTEM MODELS :

### - Context models



### - Use case Diagram:



# CHAPTER 5

- Use case text (Scenario)

## Login

|             |                                                                                                                                                                                                                                     |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Actors      | Student and supervisor and student affairs                                                                                                                                                                                          |
| Description | The system allows users to enter their credentials username along with their password, and the system must verify whether the user exists in the system's database. If the user is not found, an error message should be displayed. |
| Data        | username and password                                                                                                                                                                                                               |
| Stimulus    | the command is pressing a button                                                                                                                                                                                                    |
| Response    | The customer will be transferred to the home page                                                                                                                                                                                   |
| comment     | -                                                                                                                                                                                                                                   |

## CHAPTER 5

# Add activity ideas

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Actors      | Student                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Description | The system allows users to input ideas for activities and events. Users can select the department to which they want to submit their idea, including Entertainment Ideas Department, Scientific Ideas Department, and Collaborations Department. Once selected, users submit their idea along with a detailed description. The system is capable of storing these ideas in the software's database.                                               |
| Data        | student's selected department (Entertainment, Scientific, Collaborations)<br>Idea title and Detailed description of the idea                                                                                                                                                                                                                                                                                                                      |
| Stimulus    | User initiates the process of submitting an idea for an activity or event.                                                                                                                                                                                                                                                                                                                                                                        |
| Response    | <ol style="list-style-type: none"><li>1. The system prompts the user to select the department for their idea submission.</li><li>2. The user selects the department.</li><li>3. The system presents a form for the user to input their idea details, including the idea title and a detailed description.</li><li>4. The user fills out the form and submits the idea.</li><li>5. The system stores the idea in the software's database</li></ol> |
| comment     | This feature enables users to create content, encouraging participation and creativity in the club. Categorizing ideas into different departments makes it easier to submit ideas and ensures they are reviewed by the right people.                                                                                                                                                                                                              |

## CHAPTER 5

# Organize ideas according to priority

|             |                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Actors      | Supervisor                                                                                                                                                                                                                                                                                                                                                                                     |
| Description | <p>The system allows users to input ideas for activities and events. Users can select the department to which they want to submit their idea, including Entertainment Ideas Department, Scientific Ideas Department, and Collaborations Department. Once selected, users submit their idea along with a detailed description. The system is capable of storing these ideas in the database</p> |
| Data        | <p>User's selected department</p> <ul style="list-style-type: none"><li>- Idea title</li><li>- Detailed description of the idea</li></ul>                                                                                                                                                                                                                                                      |
| Stimulus    | <p>User initiates the process of submitting an idea for an activity or event.</p>                                                                                                                                                                                                                                                                                                              |
| Response    | <ol style="list-style-type: none"><li>1. The system prompts the user to select the department for their idea submission.</li><li>2. The user selects the department.</li><li>3. The system presents a form for the user to input their idea details, including the idea title and a detailed description.</li><li>4. The user submits the idea.</li></ol>                                      |
| comment     | <p>Storing ideas in the database facilitates easy</p> <p>This feature allows for user-generated content, fostering engagement and creativity within the club.</p>                                                                                                                                                                                                                              |

## CHAPTER 5

# Search the activity idea

| Actors      | Supervisor                                                                                                                                                                                                                                                            |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description | The system enables users to choose a category for their idea: entertainment, scientific, or collaboration. Ideas are then presented based on submission time. The club manager selects an idea, which is then presented to the supervisor along with creator details. |
| Data        | Time of idea submission                                                                                                                                                                                                                                               |
| Stimulus    | start sorting the ideas.                                                                                                                                                                                                                                              |
| Response    | <ol style="list-style-type: none"><li>1. The system displays the ideas.</li><li>2. The system allows the manager to choose the idea.</li><li>3. The idea is presented to the supervisor.</li><li>4. The system displays information about the idea creator.</li></ol> |
| comment     | This feature enables searching for ideas based on the latest submission date                                                                                                                                                                                          |



## CHAPTER 5

# Reject or accept the activity idea

|             |                                                                                                                                                                                                                                                                                                                                                                        |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Actors      | supervisor and student affairs                                                                                                                                                                                                                                                                                                                                         |
| Description | The system allows the supervisor to manage and evaluate submitted ideas based on submission time, approve or reject them, provide reasons for rejections, forward approved ideas to the dean's office, send detailed approval explanations to the dean's office, and communicate approval or rejection messages, including logical justifications, to idea submitters. |
| Data        | Acceptance or rejection                                                                                                                                                                                                                                                                                                                                                |
| Stimulus    | The supervisor begins by accepting or rejecting the idea.                                                                                                                                                                                                                                                                                                              |
| Response    | Sending acceptance or rejection to the user.                                                                                                                                                                                                                                                                                                                           |
| comment     | The advantage of this feature is providing quick responses to users within their specified time frame.                                                                                                                                                                                                                                                                 |

## CHAPTER 5

# Event Schedule

|             |                                                                                                                                                                                                                                                    |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Actors      | supervisor                                                                                                                                                                                                                                         |
| Description | The system allows the club supervisor to input confirmed events and their submission dates into a calendar within the software. Additionally, it permits sending a single message to all members containing event details and its submission date. |
| Data        | Confirmed events, Event submission dates, Event details                                                                                                                                                                                            |
| Stimulus    | The presence of approved ideas and events                                                                                                                                                                                                          |
| Response    | The dates in the calendar, the notification that will be sent to the members                                                                                                                                                                       |
| comment     | -                                                                                                                                                                                                                                                  |

## CHAPTER 5

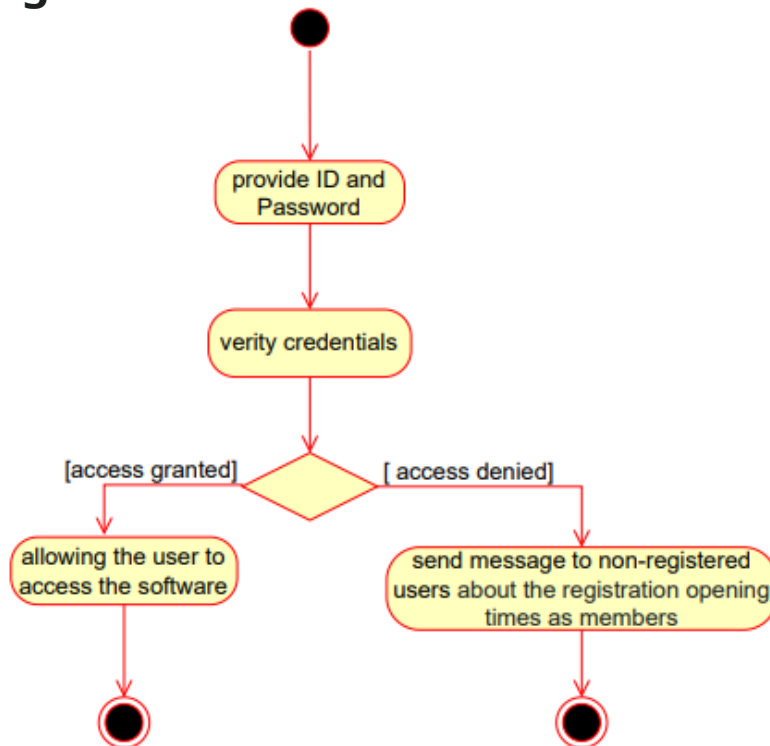
# Remind activity time

|             |                                                                                                                                                                                                                      |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Actors      | supervisor and student.                                                                                                                                                                                              |
| Description | The notification system allows for sending alerts to the club supervisor regarding the idea submitted by the member. Additionally, it enables sending a reminder message to the members 48 hours prior to the event. |
| Data        | appointment reminder,event time                                                                                                                                                                                      |
| Stimulus    | 1-There must be inputted ideas.<br>2-These ideas should be approved.                                                                                                                                                 |
| Response    | Sending notifications to the club supervisor regarding yhe idea submitted by the member, sending a reminder message to the members 48 hours before the event                                                         |
| comment     | -                                                                                                                                                                                                                    |

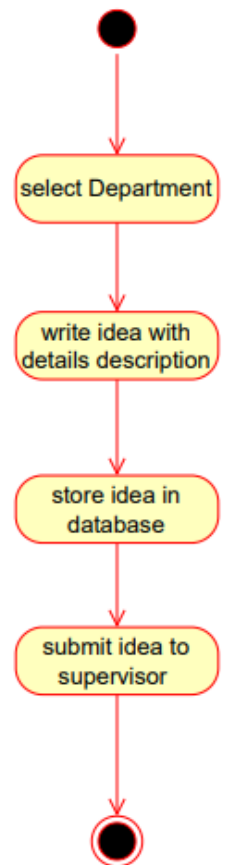
# CHAPTER 5

## - Active Diagrams

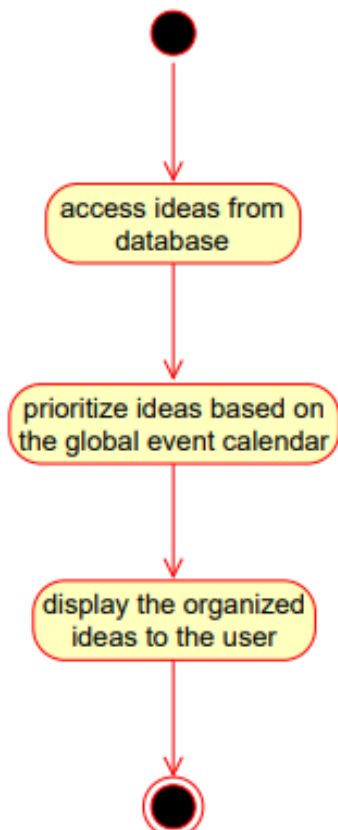
1. Login



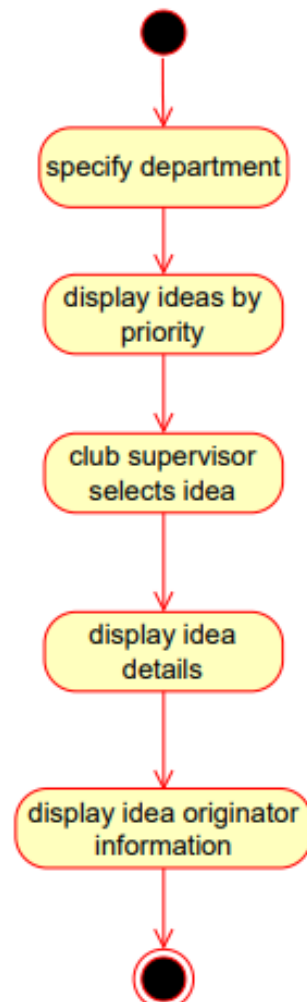
2. Add activity ideas



3. Organize ideas according to priority



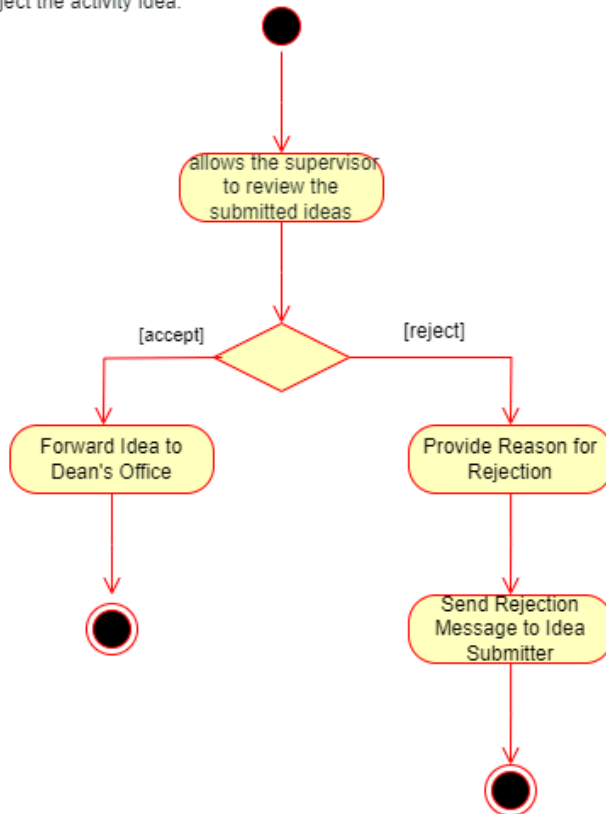
4. search the activity idea



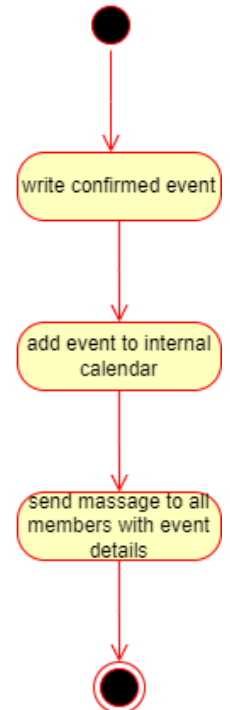
# CHAPTER 5

## - Active Diagrams

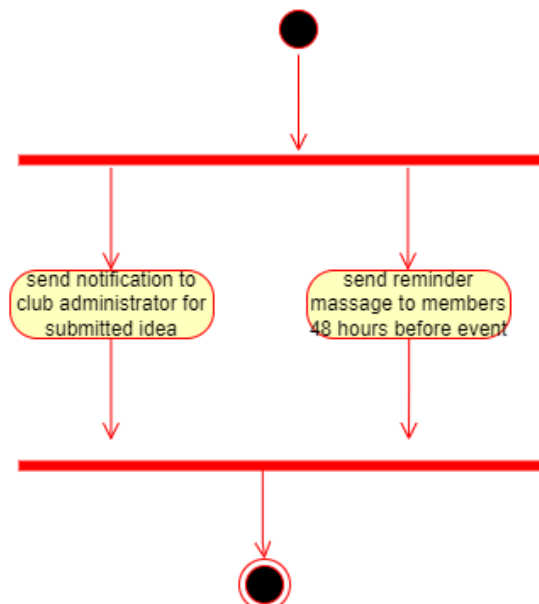
5. Supervisor can accept or reject the activity idea.



6. Scheduling



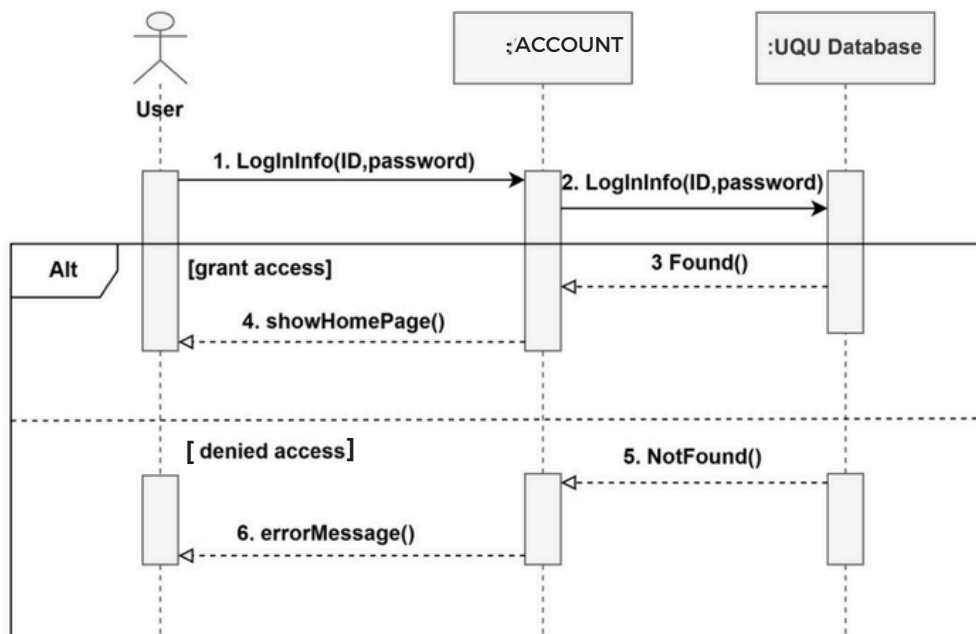
7. Reminding members and club administrator of the event date.



# CHAPTER 5

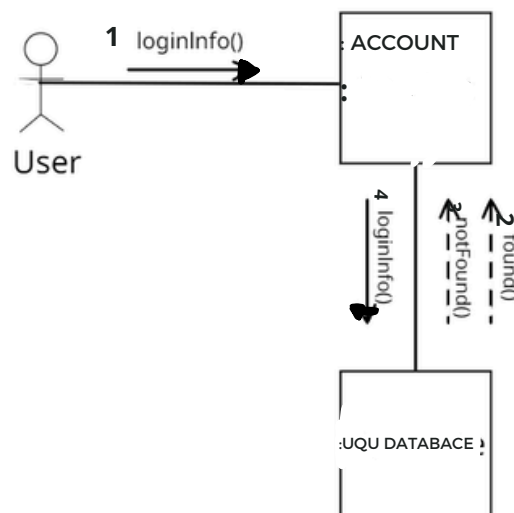
## - Sequence diagrams

1 LOG IN



## - Communication Diagram

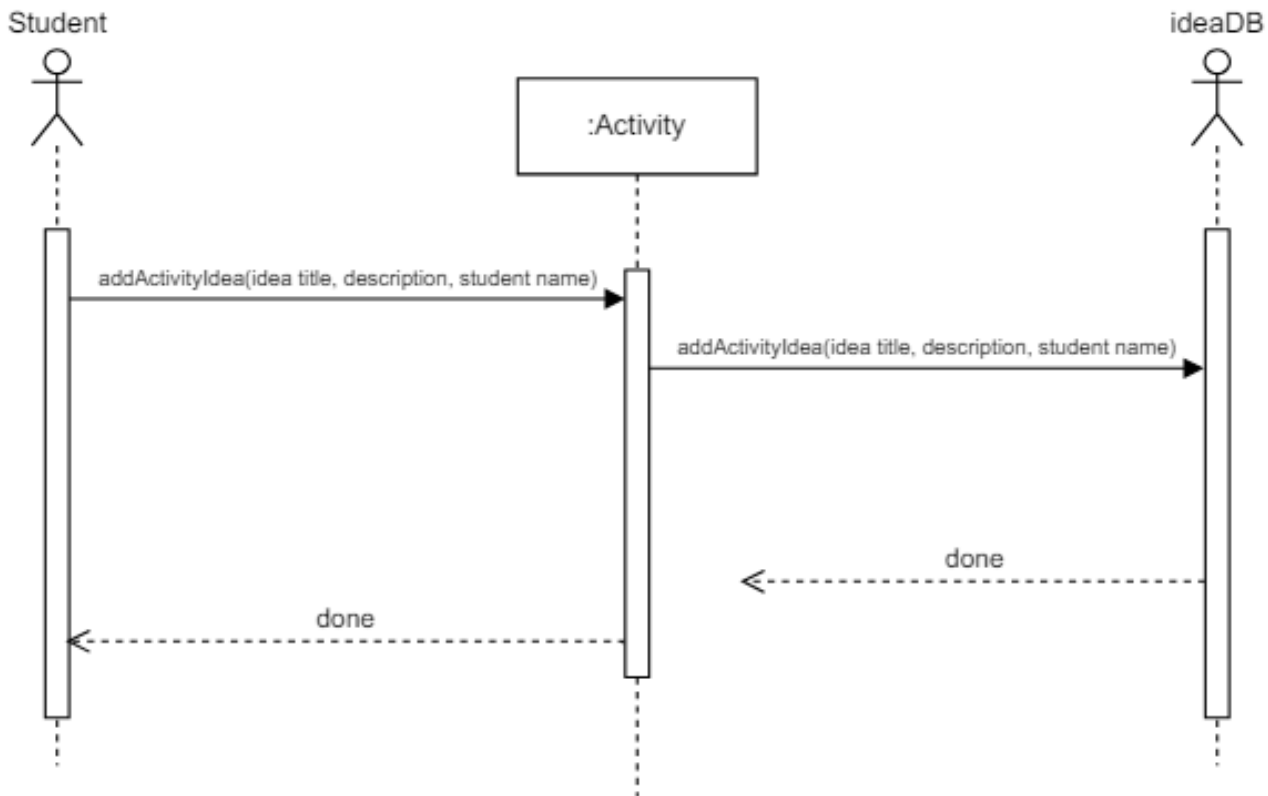
1 LOG IN



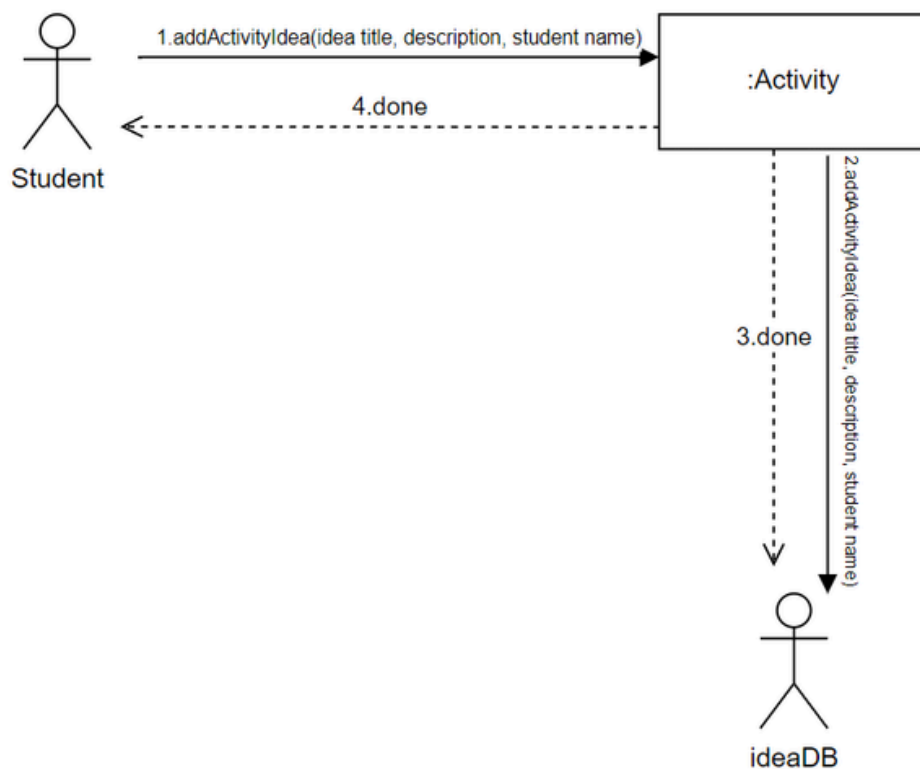
# CHAPTER 5

## - Sequence diagrams

### 2. Add activity ideas



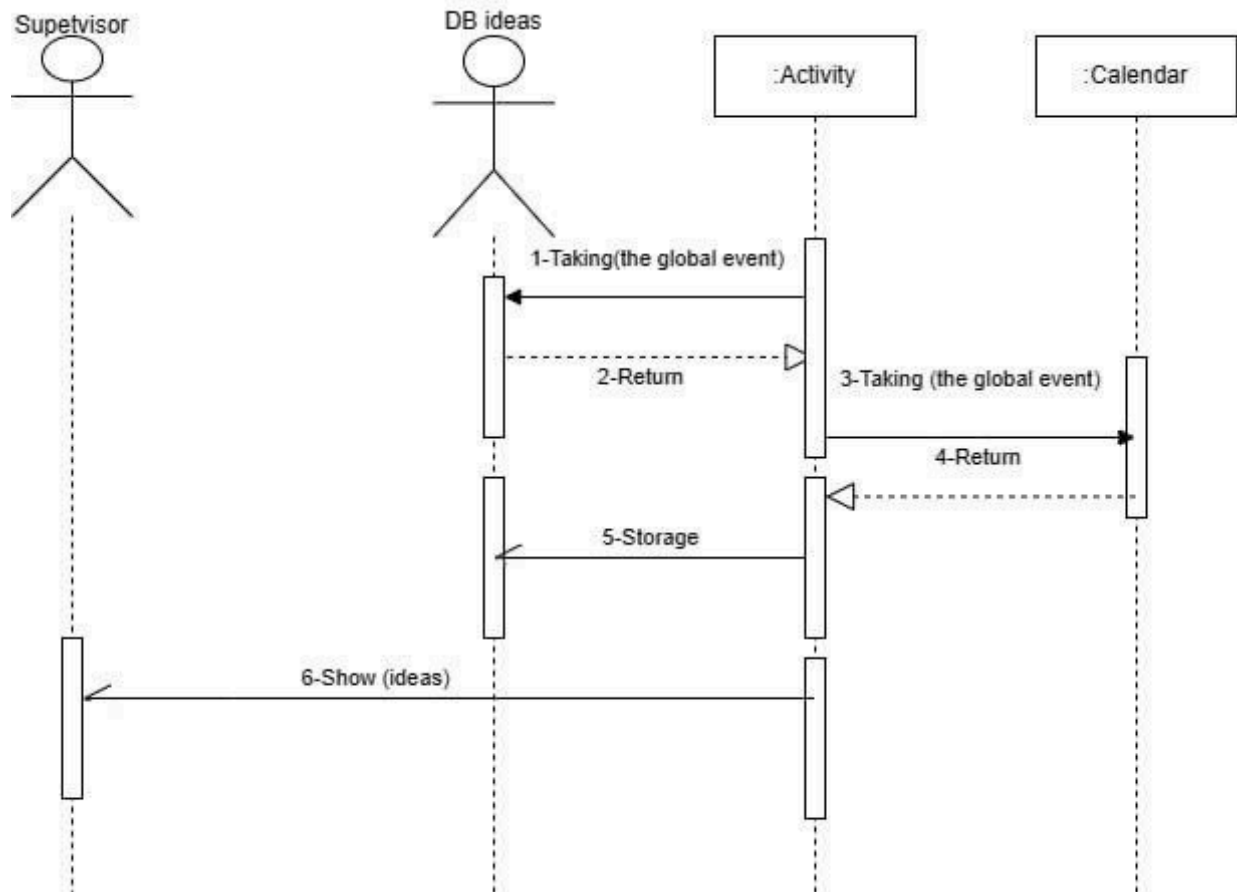
## - Communication Diagram



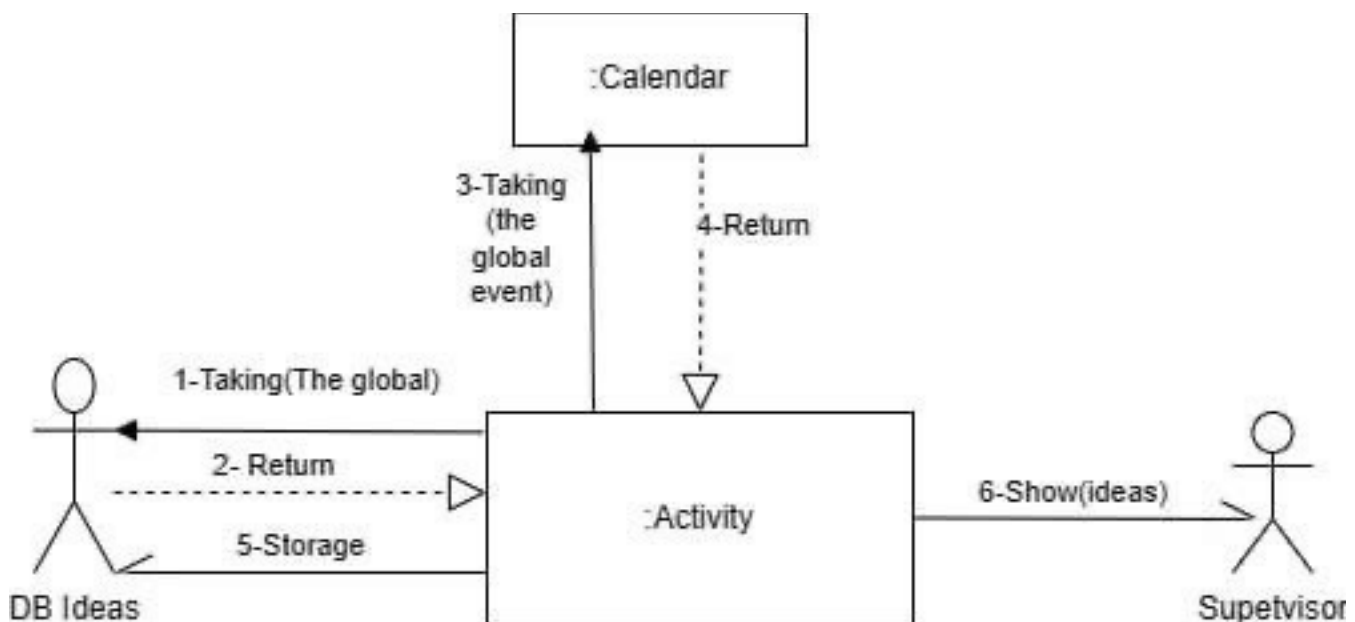
# CHAPTER 5

## - Sequence diagrams

3. Organize ideas according to priority



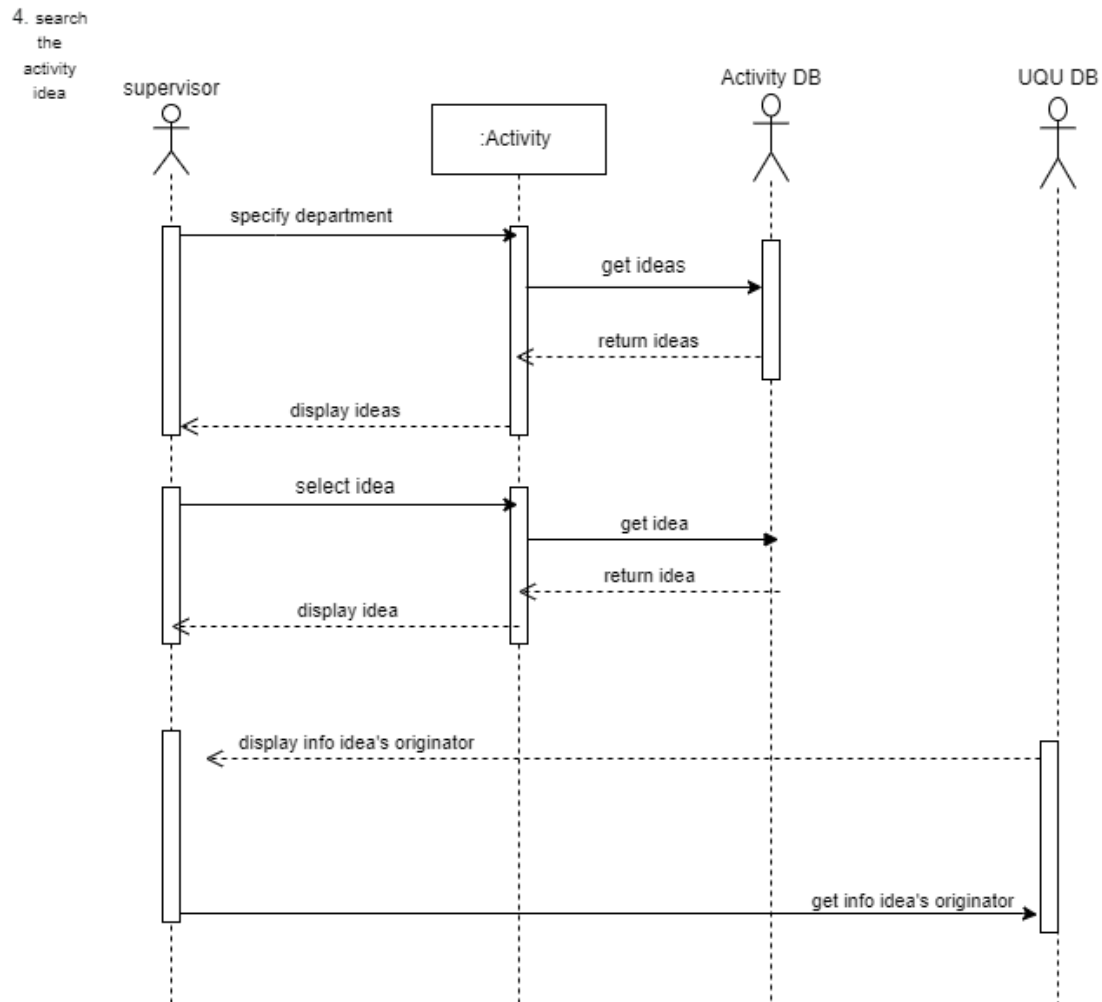
## - Communication Diagram



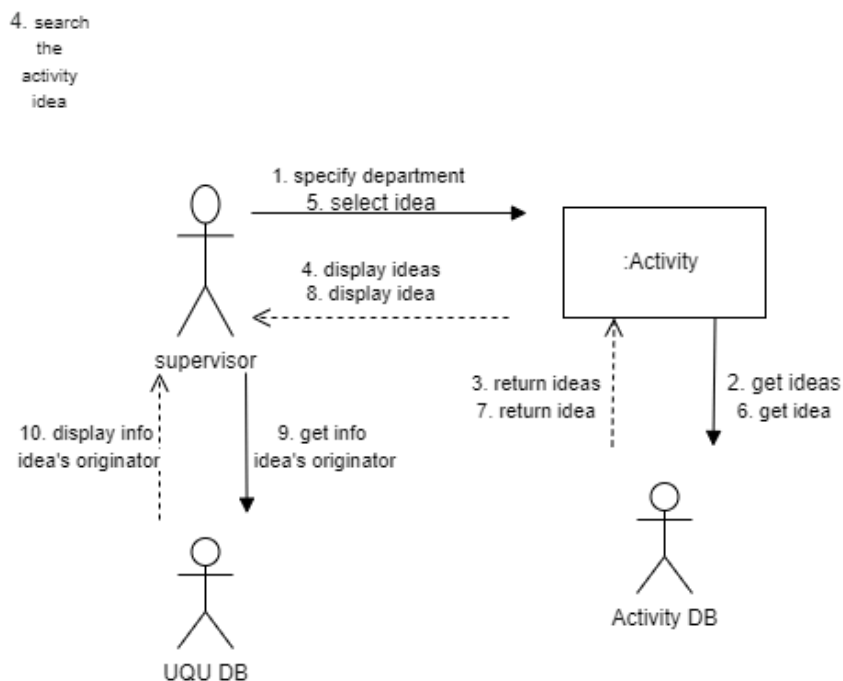


# CHAPTER 5

## - Sequence diagrams

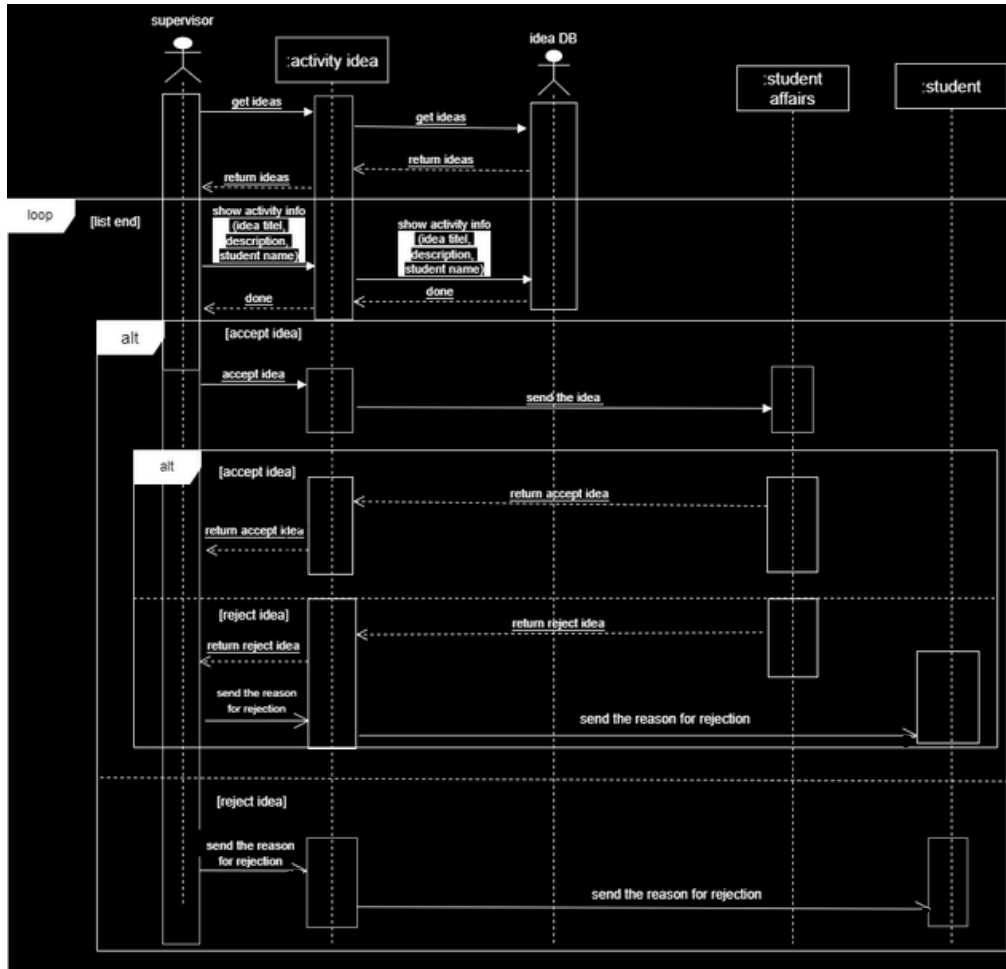


## - Communication Diagram

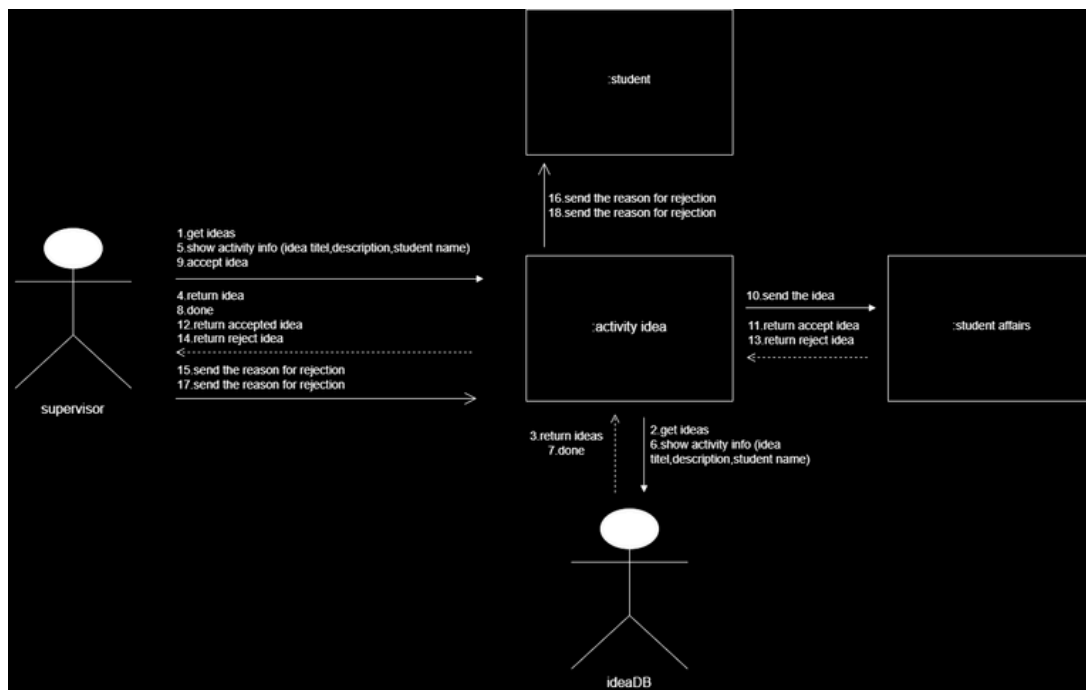


# CHAPTER 5

## - Sequence diagrams



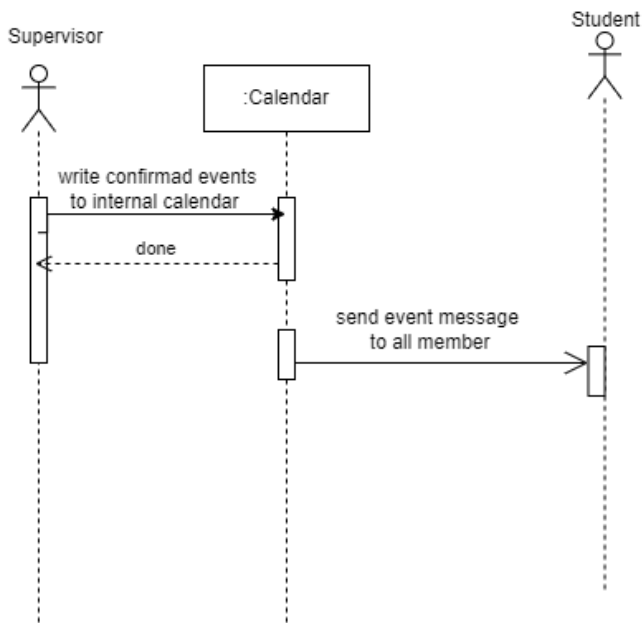
## - Communication Diagram



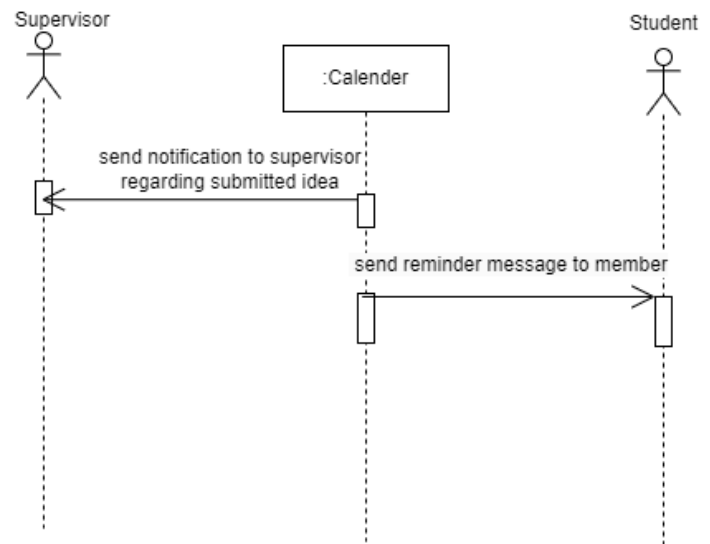
# CHAPTER 5

## - Sequence diagrams

6. Scheduling

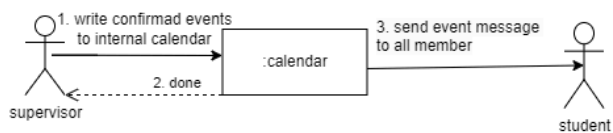


7. Remind activity time

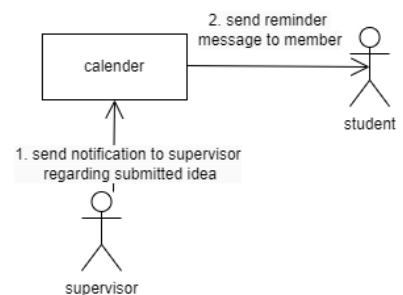


## - Communication Diagram

6. Scheduling

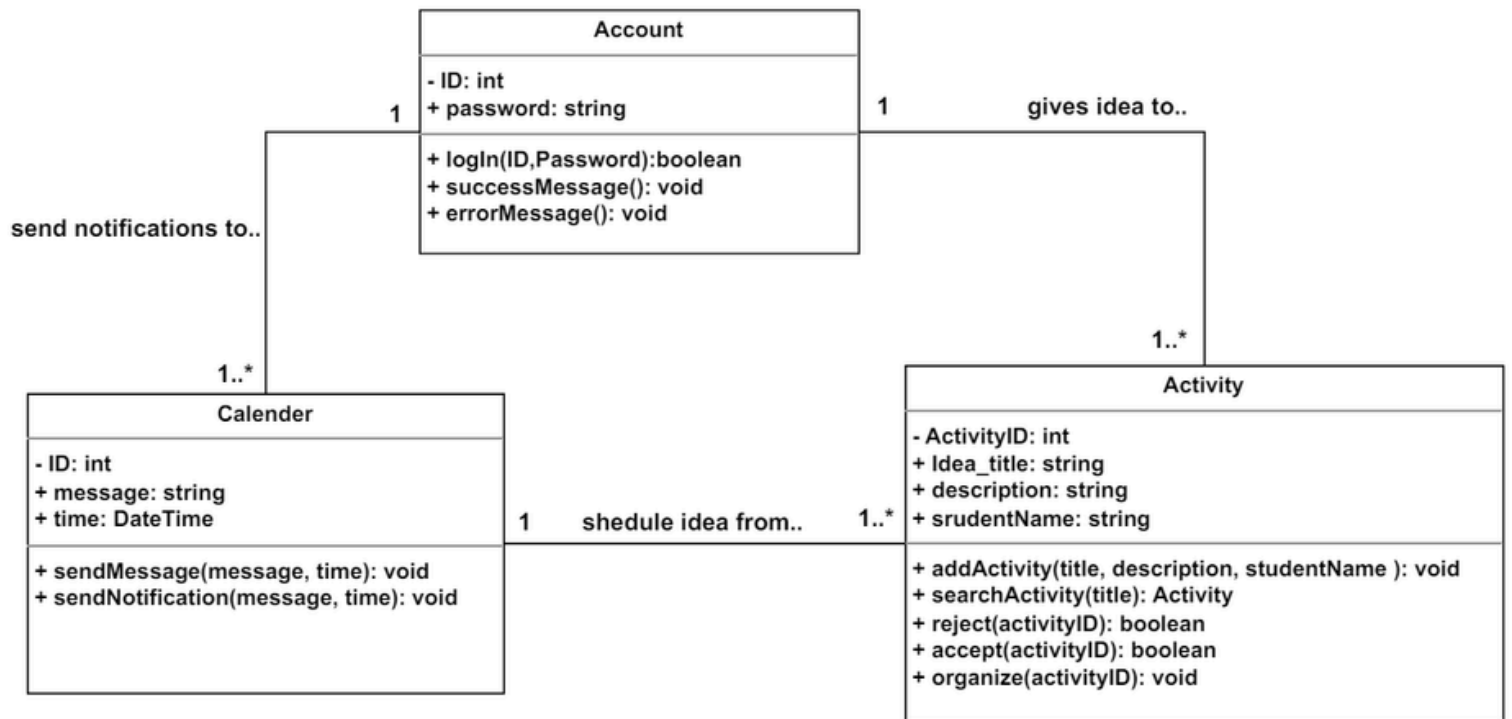


7. Reminding members and supervisor of the event date.



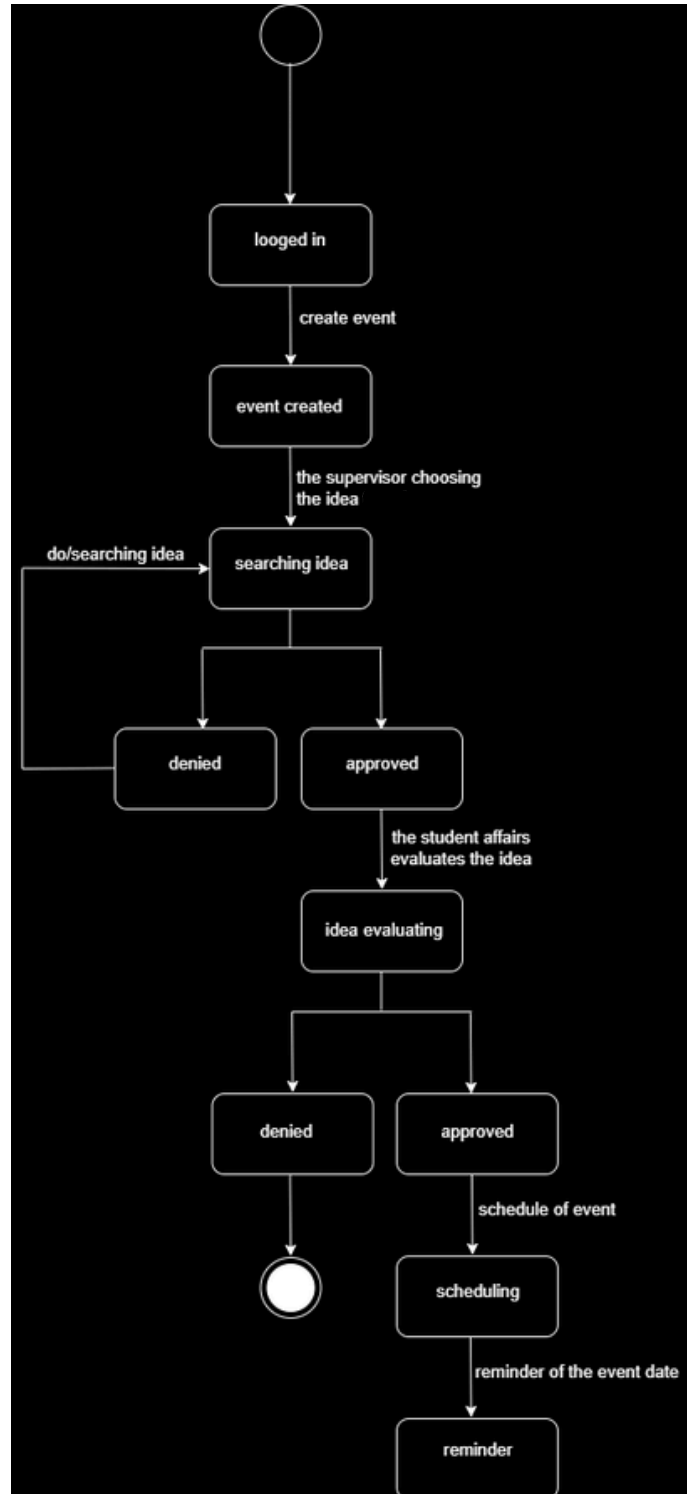
# CHAPTER 5

## - class diagram

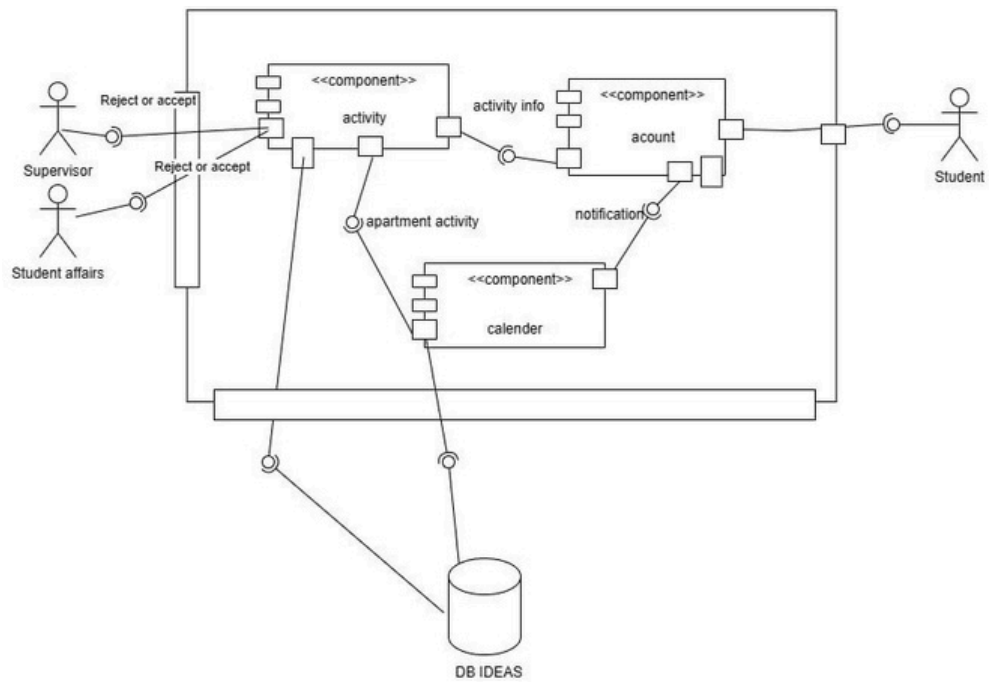


# CHAPTER 5

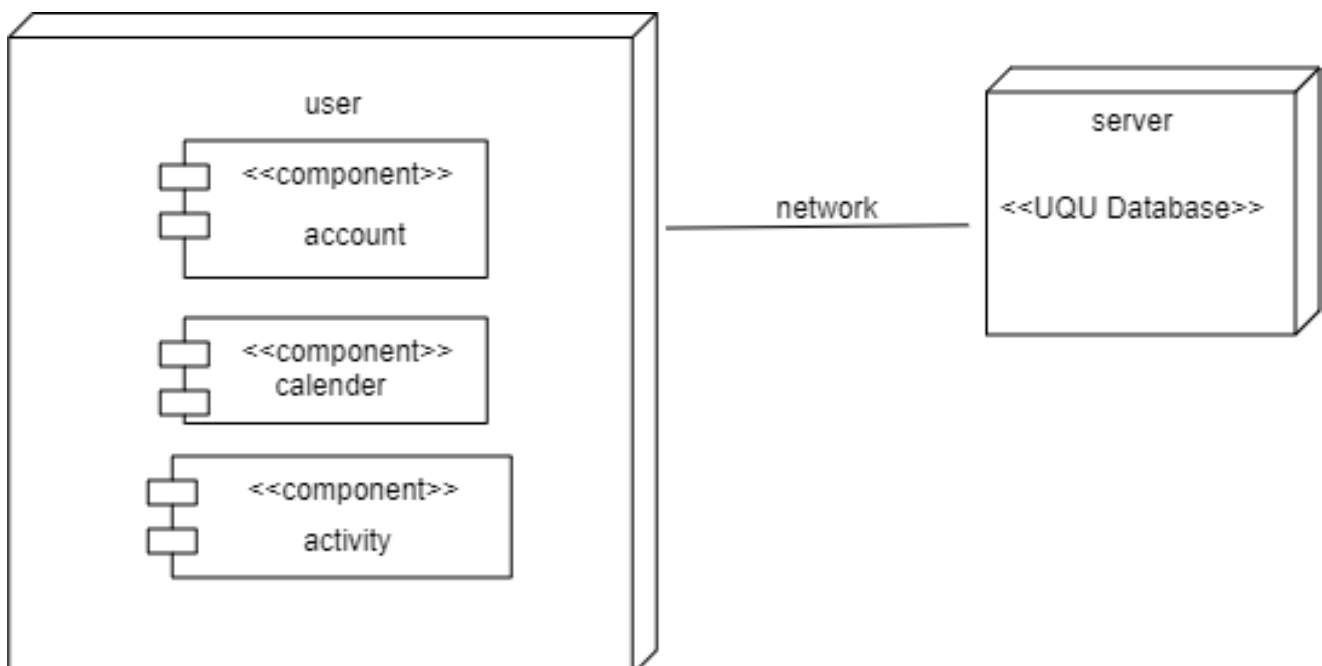
## - state diagram



## - component diagram



## - deployment diagram



## MEETING MINUETS TABLE

|   | <b>Name of group member</b> | <b>Date and Time</b> |
|---|-----------------------------|----------------------|
| 1 | arwa sultan alsulami        | 13/3/2024<br>8pm-2am |
| 2 | Noora saad alwuthainani     | 13/3/2024<br>8pm-2am |
| 3 | roba fahad almalki          | 13/3/2024<br>8pm-2am |
| 4 | jumanah alrawis             | 13/3/2024<br>8pm-2am |
| 5 | Heba Fahd Al-Matrafi        | 13/3/2024<br>8pm-2am |

## TASK TABLE

|   | <b>Name of group member</b> | <b>Task</b>                                    |
|---|-----------------------------|------------------------------------------------|
| 1 | arwa sultan alsulami        | Func requirements, non-functional requirements |
| 2 | Noora saad alwuthainani     | Func requirements, non-functional requirements |
| 3 | roba fahad almalki          | Func requirements, non-functional requirements |
| 4 | jumanah alrawis             | Func requirements, non-functional requirements |
| 5 | Heba Fahd Al-Matrafi        | Func requirements, non-functional requirements |



## TASK TABLE

|   | <b>Name of group member</b> | <b>Task</b>                         |
|---|-----------------------------|-------------------------------------|
| 1 | arwa sultan alsulami        | <b>Activity Diagrams</b>            |
| 2 | Noora saad alwuthainani     | <b>Activity Diagrams</b>            |
| 3 | roba fahad almalki          | Use case Diagram<br>Use case text - |
| 4 | jumanah alrawis             | Use case Diagram<br>Use case text - |
| 5 | Heba Fahd Al-Matrafi        | Use case Diagram<br>Use case text - |

## TASK TABLE

|   | <b>Name of group member</b> | <b>Task</b>                                        |
|---|-----------------------------|----------------------------------------------------|
| 1 | arwa sultan alsulami        | <b>Communication Diagram<br/>Sequence diagrams</b> |
| 2 | Noora saad alwuthainani     | <b>Communication Diagram<br/>Sequence diagrams</b> |
| 3 | roba fahad almalki          | <b>Communication Diagram<br/>Sequence diagrams</b> |
| 4 | jumanah alrawis             | <b>Communication Diagram<br/>Sequence diagrams</b> |
| 5 | Heba Fahd Al-Matrafi        | <b>Communication Diagram<br/>Sequence diagrams</b> |